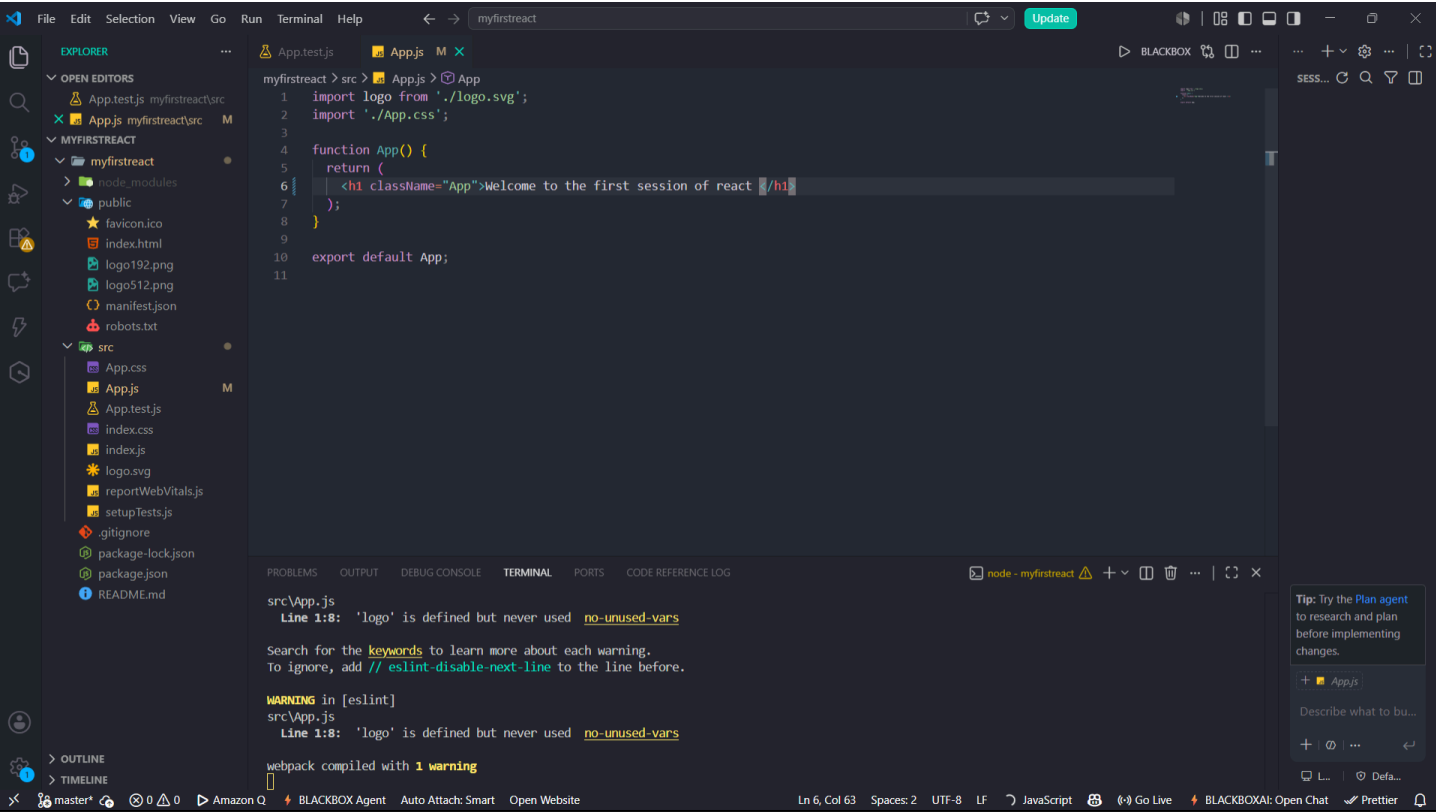
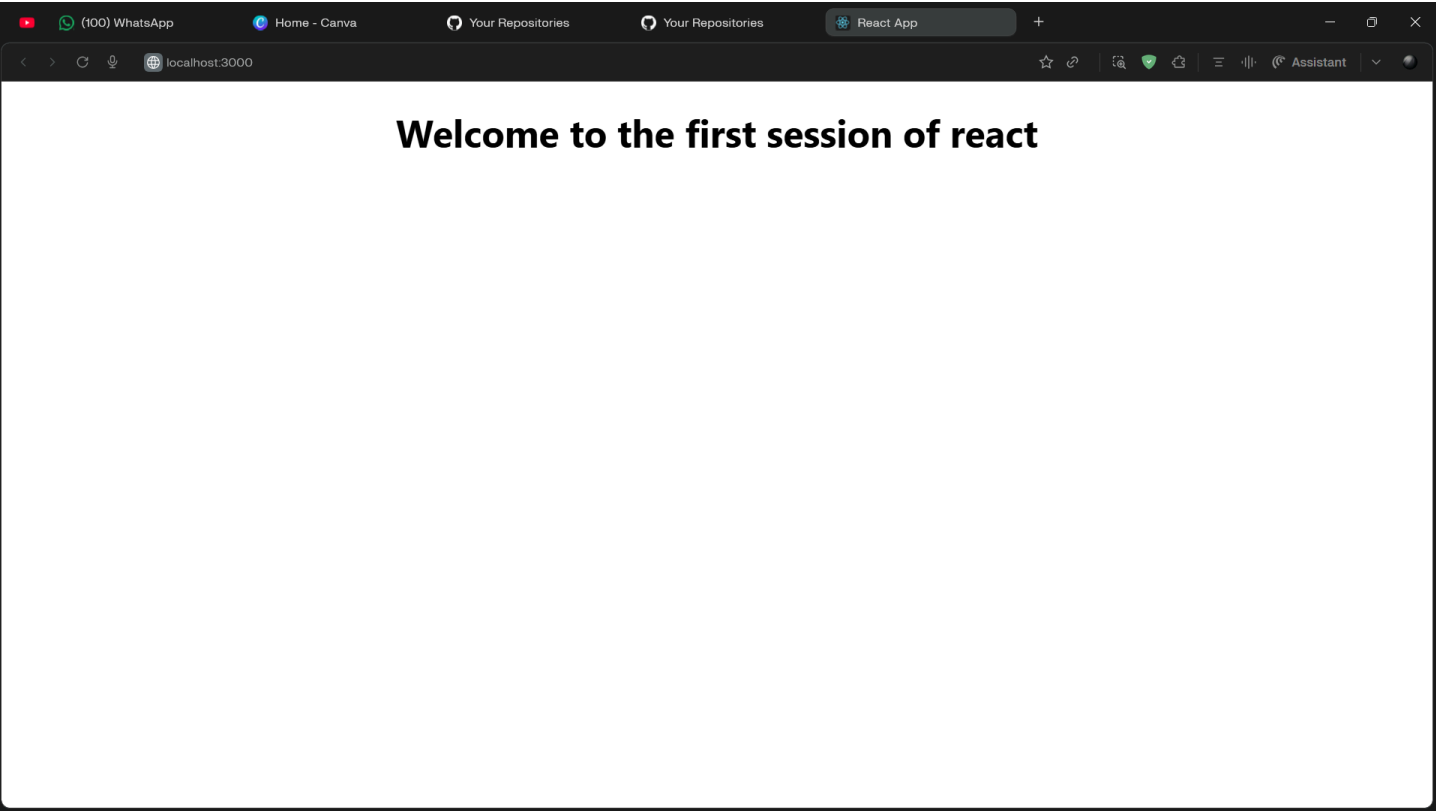


React 1

code



Output



Q1. What is a SPA and what are its benefits?

SPA means **Single-Page Application**. It is a website that loads only one HTML page. After that, when you click around, the page does not reload. Instead, JavaScript changes only the part of the page that needs to change. It feels fast, like using a mobile app.

Benefits:

- Feels fast — no waiting for full page reloads
- Less work for the server — server just sends data, not full pages
- Frontend and backend are separate, so both are easier to build and update
- Can save data on the user's device (caching) to work faster
- Same backend can be reused to build a mobile app too
- Smooth animations between screens since the page is not rebuilt each time

Q2. What is React and how does it work?

React is a free JavaScript library made by Meta (Facebook). It is used to build the visual part (UI) of websites and apps. React lets you build small, reusable pieces called **components** (like a button, a card, a navbar) and combine them to build a full page.

How it works:

- You build the UI using small components, written using JSX (HTML-like code inside JavaScript)
- When data changes, React creates a new copy of the UI in memory (called the Virtual DOM)
- React compares the new copy with the old copy to see what actually changed
- React then updates only those changed parts on the real webpage — not the whole page
- Data flows in one direction: from parent component to child component, keeping things predictable

Q3. What are the differences between SPA and MPA?

An MPA (Multi-Page Application) is a normal website where every click loads a brand-new page from the server. Here is how they compare:

Point	SPA	MPA
Page loading	One page, content updates itself	New full page loads each time
Speed after first load	Very fast	Slower, since it reloads often
First-time loading	Can be a bit slow (loads more JS)	Usually loads faster at first
Good for search engines (SEO)	Harder by default	Easier, naturally SEO-friendly
Building it	Needs more frontend logic (routing, state)	Simpler structure overall
Example sites	Gmail, Twitter/X, Trello	Wikipedia, many blogs

Q4. What are the pros and cons of a Single-Page Application?

Pros:

- Fast and smooth to use after it loads
- Better overall experience for the user
- Frontend and backend work is kept separate and clean
- Saves data usage by only fetching what's needed
- Great for building complex, app-like websites

Cons:

- Harder for search engines to read the content (SEO issues)
- First page load can be slow since it downloads a big JavaScript file
- Needs extra effort to make the back/forward browser buttons work properly
- More logic is exposed in the browser, which can bring security risks if not careful
- Needs JavaScript turned on to work at all

Q5. Can you explain more about React?

React is not a full framework — it is a library that focuses only on the UI (the part users see and click on). It is one of the most popular tools for building single-page applications today, and it is backed by Meta and a huge community of developers around the world.

Q6. What is the Virtual DOM?

The Virtual DOM is like a rough draft copy of the real webpage, kept in the computer's memory. Updating the real webpage directly is slow. So React instead:

- Makes a virtual (draft) copy of the page
- When something changes, it makes a new virtual copy
- Compares the new copy with the old one to spot the differences
- Updates only the changed parts on the real webpage

This makes React apps much faster than updating the whole page every time.

Q7. What are the features of React?

- **JSX** — lets you write HTML-like code inside JavaScript
- **Component-based** — UI is built from small, reusable pieces
- **Virtual DOM** — makes updates fast and efficient
- **One-way data flow** — data moves from parent to child, keeping things predictable
- **Hooks** — special functions like `useState` and `useEffect` to manage data and behavior
- **Declarative style** — you describe what the UI should look like, and React handles the rest
- **Big ecosystem** — extra tools like React Router (for navigation) and Redux (for managing data)
- **Strong community** — backed by Meta and used by millions of developers